

Phase 1: Requirement Analysis & Planning Document

Customer Support Ticket Priority Prediction and Automated Assignment System
Using Agentforce

Section 1: Business Requirements (Task 1)

Executive Summary & Project Objective

Modern enterprise customer support organizations face compounding operational inefficiencies driven by high ticket volumes, unstructured customer inputs, and manual triage bottlenecks. Currently, support operations rely on human dispatchers or basic rule-based scripts to review incoming requests, assess severity, assign priority labels, and route tickets to appropriate service queues. This manual intervention introduces significant latency—often adding hours to the initial response window—and creates inconsistent priority classifications due to subjective agent evaluation.

When critical system outages, security vulnerabilities, or core feature failures are submitted alongside routine inquiries, they remain queued indiscriminately until human intervention occurs. This lack of automated intelligence leads to frequent Service Level Agreement (SLA) breaches, degraded customer trust, high Support Agent cognitive fatigue, and an inability for operational leadership to gain real-time visibility into systemic product disruptions.

The primary mission of this project is to architect and deploy an enterprise-grade AI automation engine utilizing Salesforce Agentforce. By pairing generative Large Language Models (LLMs) and natural language understanding with native Salesforce Flow orchestration, the system will instantly analyze inbound ticket text, predict operational priority, assess real-time SLA risk, categorize issue types, and dynamically assign tickets to the optimal support resource without human delay.

Stakeholder Analysis

Support Agents

- **Pain Points:** Support representatives spend up to 30% of their shifts performing manual administrative triage, reading ambiguous customer descriptions, verifying customer account tiers, and manually updating object fields. Additionally, misrouted tickets require manual transfers between departments, leading to frustration and distorted personal performance metrics.
- **Business Needs:** Needs pre-classified, context-enriched tickets landed directly in their active queue. Agents require immediate visibility into key issue summaries, AI-generated reasoning for assigned priority levels, and pre-packaged action recommendations to minimize manual steps.

Support Managers

- **Pain Points:** Operational leadership lacks real-time visibility into live queue health and active capacity across tier groups. Managing SLA compliance is inherently reactive; managers discover pending breaches only after limits are exceeded. Furthermore, manual reassignment of critical tickets during high-volume incidents leads to uneven workload distribution.
- **Business Needs:** Requires real-time, automated monitoring of queue workloads, dynamic escalation mechanisms for high-risk tickets, complete auditability of AI decisions, and aggregated analytics dashboards tracking SLA adherence, MTTR, and classification accuracy.

End Customers

- **Pain Points:** End-users experience long response delays for critical issues, repeated re-explanations of issues due to multi-hop ticket transfers, and inconsistent service delivery standards.
- **Business Needs:** Demands immediate acknowledgment of critical disruptions, swift routing to qualified domain experts, transparent communication regarding expected resolution times, and rapid issue mitigation.

Workflow Analysis

Current Manual Workflow

1. **Ticket Generation:** Customer submits an issue via Email, Web Form, or Portal. The system generates a standard record with default, unverified metadata and a status of 'New'.
2. **Manual Queue Monitoring:** A Support Lead or Tier 1 Dispatcher periodically scans the central unassigned ticket queue to inspect incoming records.
3. **Manual Context Reading:** The dispatcher opens each record, reads through unstructured, multi-paragraph text descriptions, and cross-references customer account details to gauge urgency.
4. **Manual Field Population:** The dispatcher manually updates field values for Priority, Issue Category, and Routing Queue.
5. **Manual Queue Assignment:** The dispatcher evaluates active agent shifts from a static schedule and manually reassigns the record.
6. **SLA Breach Monitoring:** Passive SLA timers run in the background. If a ticket sits unaddressed due to misassignment, it breaches SLA boundaries, triggering post-facto manager escalations.

Proposed AI-Automated Workflow

1. **Ticket Generation & Event Trigger:** Customer submits a request through any channel. The platform creates a `Support_Ticket_Intelligence__c` custom record, automatically initiating an asynchronous execution event.
2. **Agentforce AI Orchestration:** Agentforce intercepts the record immediately. The underlying AI Topic ("Support Ticket Priority Analysis") processes the raw unstructured text using natural language processing.
3. **Entity & Intent Extraction:** Agentforce evaluates the text for explicit urgency signals (e.g., "urgent", "system failure", "outage", "critical"), categorizes the issue type, sets an AI Confidence Score, and computes real-time SLA risk level.
4. **Automated Record Enrichment:** Agentforce populates the `Priority__c`, `Issue_Category__c`, `SLA_Risk_Level__c`, and `AI_Confidence_Score__c` fields directly on the object.

5. **Flow-Based Dynamic Assignment:** A record-triggered Salesforce Flow fires instantly. If priority is classified as 'High' or SLA risk is elevated, the Flow bypasses standard queues, directly routes the ticket to specialized senior agents based on workload availability, creates an urgent Review Task, and notifies management via Chatter.
6. **Proactive SLA Protection:** Real-time monitoring tracks resolution timelines against calculated SLA risk, automatically rerouting or escalating records before a breach can occur.

Core System Requirements

- **Central Ticket Intake Engine:** Continuous, automated capture of unstructured customer communications across Web, Email, and API channels feeding standard Case and custom intelligence structures.
- **Automated Prioritization Engine:** An AI-driven classification pipeline capable of parsing complex unstructured narrative text, evaluating context, detecting urgency markers (e.g., "urgent", "system failure", "outage", "critical", "data loss"), and determining a standardized Priority level (High, Medium, Low).
- **Dynamic Assignment Logic:** High-efficiency routing executed via Salesforce Flow Builder that assigns records based on calculated priority, issue category, agent skill specialization, and current active queue load.
- **Proactive Escalation Framework:** Automated creation of high-priority task records and real-time manager notification triggers whenever ticket criteria indicate elevated SLA risk or low AI confidence scores.

Expected Outcomes & Metrics

- **Mean Time to Resolution (MTTR):** Achieve a minimum 40% reduction in overall MTTR across all support tiers within 90 days of deployment.
- **Customer Satisfaction (CSAT):** Increase net CSAT score by 25% due to reduced initial response times and eliminate multi-hop transfers.
- **AI Triage Accuracy:** Maintain a sustained 95% or higher accuracy rate for AI-driven Priority and Category field population.
- **Zero Critical Delay:** Ensure 0% unassigned high-priority or system-outage tickets remain in unassigned queues longer than 5 minutes post-creation.

Section 2: Project Scope & Objectives (Task 2)

In-Scope Features

- **Custom Data Model Architecture:** Design, build, and deploy the `Support_Ticket_Intelligence__c` custom object, complete with target custom fields, picklists, lookup relationships, and formula calculations.
- **Salesforce Flow Builder Automations:** Implementation of Record-Triggered Flows for record creation handling, dynamic queue and agent assignment, automated task creation, and state management.
- **Agentforce AI Integration:** Configuration of Agentforce AI Agent infrastructure, defining custom AI Topics, System Prompts, Topic Actions, guardrails, and confidence threshold configurations.
- **Automated Escalation Architecture:** Construction of real-time monitoring processes triggering automated task generation and notifications to Support Managers upon detecting SLA risk or low AI confidence.
- **Analytical Reporting & Dashboards:** Development of operational Salesforce Lightning Reports and Dashboards targeting key performance metrics, priority distributions, AI prediction accuracy, and team capacity.

Out-of-Scope Items

- **Direct Third-Party Messaging Channels:** Direct CTI/Telephony integrations, WhatsApp Business API, or external messaging platforms (scheduled for Phase 2).
- **Financial Penalty Calculations:** Complex financial contractual penalty and rebate engines for SLA breaches (handled separately by Billing/ERP systems).
- **Multi-Language Translation Engine:** Natural language translation modules for non-English ticket inputs (deferred to Phase 2 roadmap).

Measurable Objectives & KPIs

Metric / KPI	Current Baseline	Target Post-Implementation	Measurement Interval
Initial Triage Time	210 Minutes (Manual)	< 2 Minutes (Automated)	Continuous / Real-time
First Contact Resolution (FCR)	38%	55%	Monthly Audit
AI Classification Confidence	N/A (Manual)	>= 90% Average Score	Weekly AI Audit
SLA Breach Rate (High Priority)	14.2%	< 2.0%	Bi-weekly Review
Mean Time to Resolution (MTTR)	18.5 Hours	<= 11.0 Hours	Monthly Executive Report

Section 3: User Needs & Stakeholder Analysis (Task 3)

Stakeholder Mapping & Persona Breakdown

Persona 1: Tier 1 & Tier 2 Support Agents

- **Role Description:** Front-line engineers responsible for investigating, troubleshooting, and resolving customer inquiries.
- **System Interactions:** Interacts with pre-routed `Support_Ticket_Intelligence__c` records inside Lightning Console. Reviews AI summary fields, priority assignments, and executes work without performing triage.
- **Key Capabilities Required:** In-app access to AI confidence reasoning, ability to manually override priority if customer context shifts, and single-click resolution logging.

Persona 2: Support Operations Manager

- **Role Description:** Operational supervisor overseeing queue distribution, SLA compliance, capacity management, and team performance.
- **System Interactions:** Monitors live operational dashboards, receives high-priority escalation alerts, manages agent assignments, and audits low-confidence AI predictions.
- **Key Capabilities Required:** Real-time visibility into queue health, override notification hooks, access to AI performance audits, and skill-based dynamic assignment rules.

Persona 3: End Customer

- **Role Description:** External client user submitting technical queries or reporting operational disruptions.
- **System Interactions:** Submits issue details via portal forms or email; receives immediate automated updates containing confirmation and priority status.
- **Key Capabilities Required:** Rapid acknowledgment, correct immediate routing to domain experts, and predictable SLA timelines.

Persona 4: Agentforce AI Agent (Autonomous Engine)

- **Role Description:** Embedded AI reasoning engine performing natural language parsing, priority prediction, category assignment, and confidence scoring.
- **System Interactions:** Intercepts newly created ticket records, executes topic actions, populates core object fields, and hands off execution to Flow Builder.
- **Key Capabilities Required:** Natural language parsing, trigger keyword identification ("urgent", "outage"), rule enforcement, structural field output generation.

Functional Requirements Matrix

1. Ticket Management

- **FR-TM-01:** System must automatically ingest unstructured ticket text into `Ticket_Description__c` on record creation.
- **FR-TM-02:** System must maintain explicit parent-child relational links to standard `Account` and `Contact` records.
- **FR-TM-03:** System must enforce strict status state transitions across New, In Progress, Escalated, Resolved, and Closed.

2. Smart Prioritization

- **FR-SP-01:** System must evaluate incoming text for urgency keywords including "urgent", "system failure", "outage", "critical", and "data loss".
- **FR-SP-02:** System must set `Priority__c` field to High, Medium, or Low based on AI reasoning engine evaluation.
- **FR-SP-03:** System must calculate an `AI_Confidence_Score__c` percentage representing classification certainty.
- **FR-SP-04:** System must derive `SLA_Risk_Level__c` based on account tier, issue priority, and system outage status.

3. Automated Assignment & Action

- **FR-AA-01:** System must dynamically route tickets classified as 'High' priority directly to dedicated Senior Support Queues or specific active Senior Agents via Flow logic.
- **FR-AA-02:** System must automatically generate an urgent Review Task assigned to the target agent upon creation of any High-Priority record.
- **FR-AA-03:** System must trigger automated manager notification alerts via Chatter and Email whenever `SLA_Risk_Level__c` reaches 'High' or 'Breached'.

4. Monitoring & Insights

- **FR-MI-01:** System must aggregate real-time metrics showing total tickets triaged, average AI confidence, priority distributions, and SLA breach rates.

- **FR-MI-02:** System must track agent manual override events where an agent alters AI-assigned Priority or Category values for accuracy auditing.

Technical Capabilities Mapping

Functional Requirement ID	Requirement Summary	Target Salesforce Component	Architecture Role
FR-TM-01 to 03	Data Ingestion & State Management	Custom Object (Support_Ticket_Intelligence__c)	Data Storage & Lifecycle Layer
FR-SP-01 to 03	AI Priority & Intent Analysis	Agentforce AI Agent & Topic Actions	Natural Language AI Reasoning Engine
FR-SP-04	SLA Risk Assessment	Formula Fields & Agentforce Logic	Real-Time Risk Computation
FR-AA-01 to 03	Dynamic Routing & Task Creation	Salesforce Flow Builder (Record-Triggered)	Operational Process Orchestration
FR-MI-01 to 02	Analytics & Manual Override Auditing	Lightning Reports & Dashboards	Management Visibility & Auditing

Section 4: Salesforce Features & Tool Architecture (Task 4)

Core System Architecture

The architecture follows an event-driven, decoupled design leveraging native Salesforce platform capabilities across three primary functional layers:

1. **Data Storage Layer:** Enterprise metadata and operational records reside within the custom object `Support_Ticket_Intelligence__c`, linked via

standard Lookups to **Account**, **Contact**, and **User** (Assigned Agent) objects.

2. **AI Orchestration Layer (Agentforce)**: Operates on incoming data asynchronously. Intercepts newly created **Support_Ticket_Intelligence__c** records, passes the **Ticket_Description__c** to the LLM agent, executes topic instructions, populates predicted metadata, and outputs the **AI_Confidence_Score__c**.
3. **Process Orchestration Layer (Flow Builder)**: Triggered immediately following Agentforce record field updates. Evaluates execution paths via decision trees, manages agent capacity assignment, creates downstream **Task** records, and handles multi-channel alerting.

Flow & Automation Logic

Input / Output Variables

- **Input Variables:** **Record.Id**, **Record.Ticket_Description__c**, **Record.Account__c**, **Record.Contact__c**.
- **Output Variables:** **Record.Priority__c**, **Record.Issue_Category__c**, **Record.Assigned_Agent__c**, **Record.SLA_Risk_Level__c**, **Record.Status__c**.

Decision Tree Branching Logic

[Record Created: Support_Ticket_Intelligence__c]

|



[Agentforce AI Priority Analysis Executed]

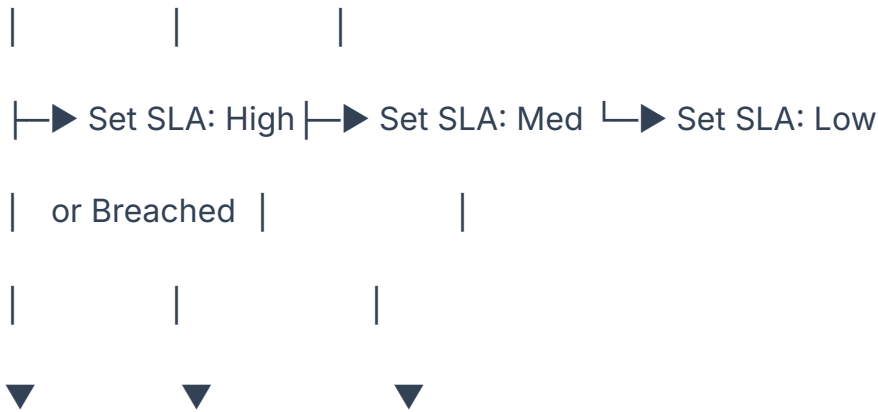
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[Decision: Priority Classification]



[High Priority] [Medium Priority] [Low Priority]



[Assign to Tier 3][Assign to Tier 2][Assign to Tier 1]

Senior Queue] General Queue] Standard Queue]



[Create Urgent [Create Standard [Create Low Priority

Task for Agent] Task for Queue] Task for Queue]



[Send Chatter &

Email Alert to

Support Manager]

Detailed Rule Set

- **Branch 1 (Urgent / System Outage):** If `Priority__c` = 'High' OR `Issue_Category__c` = 'System Outage' OR text contains critical keywords ("system failure", "outage", "critical"), set `SLA_Risk_Level__c` = 'High', route directly to Senior Tier 3 Support Queue, generate a High-Priority `Task` assigned to the designated lead, and fire a real-time Chatter alert to the Support Manager.
- **Branch 2 (Standard Technical Issue):** If `Priority__c` = 'Medium', set `SLA_Risk_Level__c` = 'Medium', route to Tier 2 Technical Queue via Round-Robin load distribution.
- **Branch 3 (General Inquiry):** If `Priority__c` = 'Low', set `SLA_Risk_Level__c` = 'Low', route to Tier 1 General Support Queue.

Agentforce AI Configuration

AI Topic Specification

- **Topic Name:** Support Ticket Priority Analysis
- **Developer Name:** `Support_Ticket_Priority_Analysis`
- **Classification Scope:** Evaluates customer support ticket narrative text to predict issue urgency, operational category, and classification confidence score.

System Prompt & Instructions

You are an expert enterprise support triage agent. Your objective is to analyze raw ticket narrative descriptions (`Ticket_Description__c`) and extract structured classifications. Instructions:

1. Read the ticket text carefully. Scan for explicit critical failure terms including: "urgent", "system failure", "outage", "critical", "down", "data loss", or "security breach".
2. Assign Priority (**Priority__c**):
 - Set to 'High' if critical keywords are present, or if the user indicates a complete business stoppage or core function failure.
 - Set to 'Medium' if the issue relates to performance degradation, isolated feature bugs, or non-blocking account access issues.
 - Set to 'Low' for general inquiries, how-to questions, minor UI bugs, or feature requests.
3. Categorize Issue (**Issue_Category__c**): Map the inquiry to 'System Outage', 'Performance', 'Account Access', or 'General Inquiry'.
4. Evaluate Certainty: Output an **AI_Confidence_Score__c** between 0.00 and 1.00 reflecting your classification certainty based on text clarity.

Topic Actions

- **Action Name:** **Update_Ticket_Intelligence_Record**
- **Action Logic:** Binds generated LLM field outputs directly to the triggering **Support_Ticket_Intelligence__c** record ID.

Execution Triggers

Invoked asynchronously via platform event immediately upon creation of a **Support_Ticket_Intelligence__c** record where **Status__c** = 'New'.

Security & Access Control Strategy

Profile Configurations

- **Support Agent Profile:** Read/Edit access to assigned **Support_Ticket_Intelligence__c** records. Read-only access to **AI_Confidence_Score__c** and **SLA_Risk_Level__c**.
- **Support Manager Profile:** Full Read/Create/Edit/Delete access to all records across all queues. Full override capabilities.

- **System Administrator Profile:** Full administrative access, schema management, flow maintenance, and Agentforce prompt editing.

Permission Sets

- **Agentforce_AI_Admin_PS:** Grants full administrative control over Agentforce AI Topics, Prompt Templates, and Action definitions.
- **Support_Intelligence_User_PS:** Grants operational end-users access to the `Support_Ticket_Intelligence__c` custom object, Lightning Console apps, and relevant execution flows.

Object & Field-Level Security (FLS)

- **Object-Level Access:** Restricted via Profile and Permission Sets. Default global access set to strict Private.
- **Field-Level Security:** `AI_Confidence_Score__c` and `SLA_Risk_Level__c` are set to Read-Only for standard Support Agent profiles to prevent manual tampering with system-computed metrics while allowing Support Managers full Read/Edit capabilities.

Section 5: Data Model & Security Model Specification (Task 5)

Custom Object Specification

- **Object Label:** Support Ticket Intelligence
- **Plural Label:** Support Ticket Intelligences
- **Object API Name:** `Support_Ticket_Intelligence__c`
- **Description:** Central custom object storing unstructured customer ticket inputs, AI-generated priority predictions, issue categorizations, calculated SLA risk scores, confidence metrics, and dynamic agent routing assignments.

Custom Fields Table

Field Label	API Name	Data Type	Length / Picklist Values	Description
Ticket Description	Ticket_Description__c	Long Text Area ▾	32,768	Raw, unstructured customer inquiry text ingested from support channels.
Priority	Priority__c	Picklist ▾	HighMediumLow	Predicted operational ticket priority assigned by Agentforce AI or manual override.
Status	Status__c	Picklist ▾	NewIn ProgressEscalatedResolvedClosed	Lifecycle state of the support ticket record.
Issue Category	Issue_Category__c	Picklist ▾	System OutagePerformanceAccount AccessGeneral Inquiry	Categorized domain of the customer issue derived by AI parsing.
SLA Risk Level	SLA_Risk_Level__c	Picklist ▾	LowMediumHighBreached	Calculated real-time risk level of breaching contractual SLA deadlines.
AI Confidence Score	AI_Confidence_Score__c	Percent ▾	3 Digits, 2 Decimals	Certainty metric (0.00% - 100.00%) generated by Agentforce AI during classification.
Assigned Agent	Assigned_Agent__c	Lookup ▾	Lookup(User)	Relational lookup link to the specific internal Support User assigned to the ticket.

Field Label	API Name	Data Type	Length / Picklist Values	Description
Resolution Time Hours	Resolution_Time_Hours__c	Number ▾	18 Digits, 2 Decimals	Total elapsed operational hours calculated upon ticket reaching 'Resolved' or 'Closed' status.

Object Relationships

- **Account__c (Lookup to Account):** Links the ticket intelligence record to the customer's enterprise Account record for SLA tier derivation.
- **Contact__c (Lookup to Contact):** Links the ticket intelligence record to the individual customer Contact reporting the issue.
- **Assigned_Agent__c (Lookup to User):** Identifies the active Salesforce User (Support Engineer) assigned to resolve the ticket.

Security Model Details

Organization-Wide Defaults (OWD)

- **Default Internal Access:** Private
- **Default External Access:** Private

Role Hierarchy Structure

- **Level 1 (Top):** VP of Customer Support (Full access to all records across the organization).
- **Level 2:** Support Operations Managers (Read/Edit/Delete access to all records within their regional team branch).
- **Level 3:** Tier 1 / Tier 2 / Tier 3 Support Agents (Read/Edit access restricted to records owned by the user or shared via queue membership).

Sharing Rules

- **Rule Name:** High_Priority_Escalation_Sharing
- **Rule Type:** Criteria-Based

- **Criteria:** `Priority__c` EQUALS 'High' OR `SLA_Risk_Level__c` EQUALS 'High', 'Breached'.
- **Shared With:** Role and Subordinates → Support Operations Managers.
- **Access Level:** Read/Write.

Field-Level Security (FLS) Enforcement Matrix

Field API Name	Support Agent Profile	Support Manager Profile	System Administrator
<code>Ticket_Description__c</code>	Read / Edit ▾	Read / Edit ▾	Read / Edit ▾
<code>Priority__c</code>	Read / Edit ▾	Read / Edit ▾	Read / Edit ▾
<code>Status__c</code>	Read / Edit ▾	Read / Edit ▾	Read / Edit ▾
<code>Issue_Category__c</code>	Read / Edit ▾	Read / Edit ▾	Read / Edit ▾
<code>SLA_Risk_Level__c</code>	Read Only ▾	Read / Edit ▾	Read / Edit ▾
<code>AI_Confidence_Score__c</code>	Read Only ▾	Read / Edit ▾	Read / Edit ▾
<code>Assigned_Agent__c</code>	Read Only ▾	Read / Edit ▾	Read / Edit ▾
<code>Resolution_Time_Hours__c</code>	Read Only ▾	Read Only ▾	Read / Edit ▾